Clustering Algorithms For Regression-Like Modeling

Applications in Market Segmentation & Predictive Maintenance

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Abstract

In today's data-driven world, organisations are constantly seeking innovative ways to extract insights from their data and drive business value. Unsupervised learning approaches, a subset of machine learning, offer a powerful tool for uncovering hidden patterns and relationships in data without prior knowledge of the expected outcome. As such, this report presents two studies on the application of unsupervised learning approaches for digital market segmentation and predictive maintenance. The first study uses Fuzzy C-Means to estimate the digital maturity of airlines, while the second study utilises Self-Organizing Maps (SOM) to predict the degradation of assets.

Keywords: Unsupervised Learning, Digital Maturity, Predictive Maintenance, Fuzzy C-Means, Self-Organizing Map

1. Introduction

This report presents two examples of using unsupervised learning approaches for regression-like modelling, with applications in digital market segmentation and predictive maintenance. The first study employs Fuzzy C-Means (FCM) to estimate the digital maturity of airlines, while the second study utilises Self-Organizing Maps (SOM) for predictive maintenance.

This report is organised in six sections. Following this introduction, a glossary will be presented in Section 2. The internship scope and rationale behind it will be presented in Section 3. Section 4 will present the key principles and results of the digital market segmentation, while Section 5 will detail the predictive maintenance study. Finally, the paper will be concluded in Section 6.

2. Glossary

Term	Definition
A/C	Aircraft
AOG	Aircraft-On-Ground
CBM	Condition-Based Maintenance
DL	Deep Learning
DMI	Digital Maturity Index
FCM	Fuzzy C-Means
ICAO	Code designating an airline or a leasing company (e.g. "AFR" for Air France)
IT	Information Technology
MAE	Mean Absolute Error
ML	Machine Learning
PdM	Predictive Maintenance
RMSE	Root Mean Square Error
SOM	Self-Organizing Map

Table 1:*Glossary*

3. Context & Key Issues

3.1. Airbus

Airbus is a multinational aerospace manufacturer that develops, manufactures, supplies, and supports a wide range of aerospace products and services. Although headquartered in Europe, Airbus employs more than 154,000 people worldwide and has offices all over the world [1].

Airbus is divided into three main divisions:

- Airbus Commercial Aircraft: Headquartered in Toulouse, France, it is in charge of civil aircraft platforms (e.g., passenger aircraft, freighter aircraft, corporate jets) and services (e.g., in-service support, data analytics, maintenance services).
- Airbus Defense and Space: Headquartered in Munich, Germany, it is in charge of military aeronautical platforms (e.g., fighter aircraft, tankers), space systems (e.g., satellites, launchers, rovers), and security services (e.g., intelligence services, cybersecurity solutions).
- Airbus Helicopters: Headquartered in Marignane, France, it is in charge of both civil and military helicopter platforms.

Airbus Commercial Aircraft, the largest of the three divisions, currently produces four families of commercial aircraft - the A320 family, A330 family, A350 family and A220 family - and continues to support them as well as legacy programs such as the A380 [2].

3.2. Scope

The original scope of this study was to investigate the digital maturity of airlines, with the aim of developing a comprehensive framework for assessing digital status of each airline. However, and despite efforts to obtain industry-wide data, it became clear that the data required to conduct a thorough analysis of digital maturity across the airline industry would not be available within the timeframe of this apprenticeship. As a result, the study was refocused to examine the digital maturity of Skywise's current customers.

As the study's scope was narrower than initially intended, it was decided to conduct an additional study on predictive maintenance to ensure a meaningful utilisation of the apprenticeship period. This complementary study aimed to explore the potential of data science in improving maintenance scheduling, and its findings are presented in a separate section of this report.

3.3. Skywise

Skywise is Airbus' big data data platform that both the aircraft manufacturer and its customer airlines use to analyse operational data and make better-informed decisions. Initially established in 2015 for internal use cases only, Skywise has been opened up to worldwide airlines in 2017, during the Paris Air Show, as a PaaS offering (Platform-as-a-Service).

Since then, the platform has grown to over 47,000 MAU (monthly active users), servicing more than 150 airlines and hosting data related to over 12,000 aircraft worldwide (little over 50% represent Airbus manufactured aircraft, the rest being from other manufacturers).

The Skywise PaaS offering provides customers with dozens of state-of-the-art tools to manipulate, analyse, visualise and transform their data, thus supporting them in their digital transformation journey. However, on top of the platform, Airbus also provides off-the-shelf solutions that directly address airline pain points, without the need for customers to write any code. One of the most popular solutions is Skywise Fleet Performance, which specialises in predictive maintenance and provides customers with timely alerts that allow them to foresee maintenance operations and avoid operational interruptions. Solutions like Fleet Performance constitute the SaaS offering of Skywise (Software-as-a-Service).

3.4. Zero AOG

The second study is performed within the Zero AOG team in Toulouse; one of four plateaus located across four different sites: Toulouse, Filton, Hamburg and Getafe whose mission is to minimise aircraft downtime and optimise fleet availability.

The team brings together experts from Customer Services and Engineering to improve aircraft maintenance. They deliver solutions for support and services around Health Monitoring and work to improve system design, support customer care, and collaborate with data analytics teams across sites.

4. Digital Maturity of Airlines

4.1. Objectives

Until today, Airbus's understanding of the digital solutions market has been based mainly on external studies coming from various consulting firms and direct qualitative feedback from select customers. While these studies are insightful, they are not sustainable over time due to the lack of access to the primary granular data behind them. On the other hand, the customer feedback is not always comprehensive and/or objective enough and it certainly doesn't shed any light on the needs of airlines who are not yet Airbus customers. As such, the overall objective of this part of the apprenticeship is to develop a generic, data-driven and internally-mastered method to analyse the digital maturity status of airlines in order to better understand the digital solution market within the aviation industry.

The original objective of this study was to conduct a comprehensive, industry-wide analysis of digital maturity. However, due to unforeseen challenges in obtaining the necessary data, the scope of the study had to be adjusted. As a result, this report will primarily focus on assessing the digital maturity of Skywise's current customers with the following sub-objectives:

1. Define key metrics and indicators to be considered in assessing digital maturity of an airline.
2. Develop a method to identify airlines with similar digital profiles.
3. Translate these groupings of airlines into formal digital maturity status.

The following sections will present in greater detail the implementation of the above sub-objectives.

4.2. State of the Art

In today's complex and competitive marketing environment, it is crucial for businesses to develop effective marketing strategies to stay ahead of the curve. The marketing literature consistently emphasises the significance of segmentation, targeting, and positioning in marketing strategies [3].

Since the scope of this study is the market of digital solutions, the state-of-the-art section of the report will focus on two main areas: (1) understanding digital maturity and how to evaluate it, and (2) benchmarking market segmentation techniques.

4.2.1. Digital Maturity

4.2.1.1. Definitions

Developing an understanding of digital maturity requires a thorough comprehension of the core concepts that underpin it, such as digital, digitalisation, digitisation, digital transformation, and then digital maturity itself. This includes the similarities and differences in their definitions, thereby providing a solid foundation for understanding the nuances of digital maturity [4].

In the Garter [5] definition, the term digital refers to something that uses or is related to digital technology, which is based on the use of binary code to process and store information. In a broader sense, "digital" can also refer to anything that is presented in a discrete, numerical format, as opposed to an analog format.

Also according to Gartner [5] , digitisation is the process of converting information or data from a physical or analog format into a digital format without changing the process itself. On the other hand, digitalisation is defined as the process of using digital technologies to change a business model, process, or system. It involves using digital tools and platforms to create new value propositions, improve efficiency, and reduce costs.

Digital transformation is a broader concept that involves using digital technologies to fundamentally change the way an organisation operates, including its business model, products, services, and customer interactions [6].

This concept of responding to digital trends is not well expressed by the word “transformation” as the response is generally learned rather than instinctive, making the word maturity (in the psychological sense) more suitable [4]. Thus, digital transformation is a process that can in theory result in digital maturity. A digital mature company is one that has finished its transformation phase and is actively using quantitative measures and data analytics to make informed decisions.

4.2.1.2. Digital Maturity Models

The objective of this section is to benchmark frameworks assessing digital maturity across different industries and that might be applicable to Airbus’s Digital Solutions.

4.2.1.2.1. Deloitte/TM Forum Model

Deloitte has developed a Digital Maturity model to support digital transformation projects in the telecommunication sector in partnership with the TM Forum, an organisation grouping together digital service providers, technology suppliers, consultancies and system integrators [6].

This DMM evaluates digital capabilities across five clearly defined business dimensions to create a holistic view of digital maturity across organisations. The core five dimensions are divided into 28 sub-dimensions, which in turn breakdown to 179 individual criteria on which digital maturity is assessed. The 28 sub-dimensions are shown in figure 1 below; and details of the individual criteria are however not disclosed:

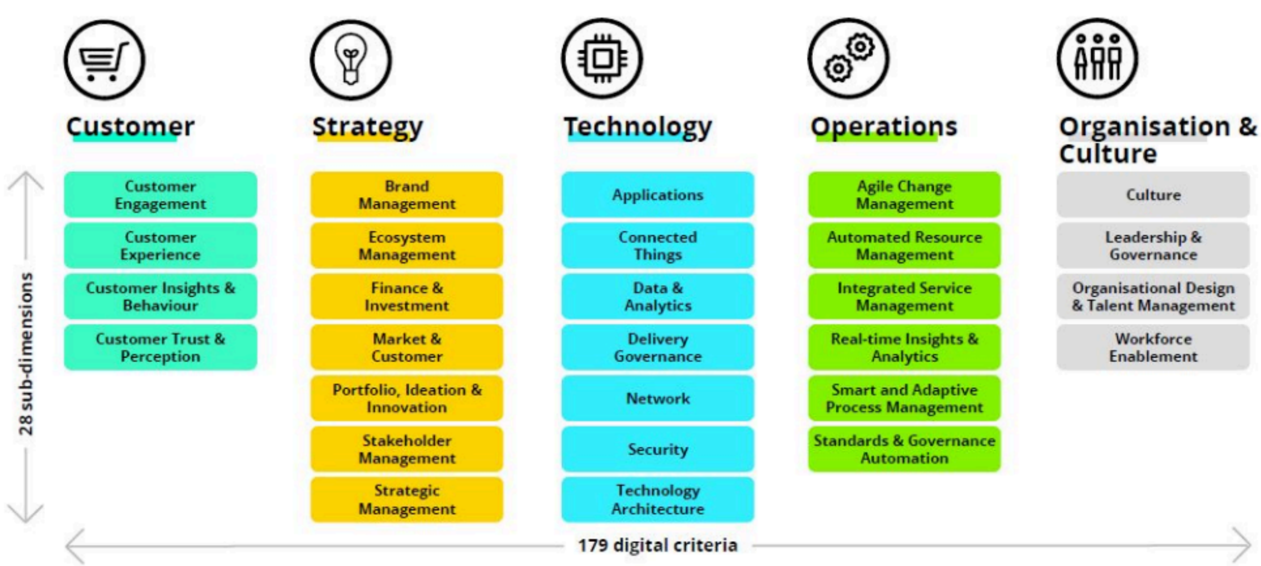

***Figure 1**: Criteria of the Deloitte/TM Forum Model*

4.2.1.2.2.Gartner Model

The Gartner consulting firm has developed several digital maturity models, specific per domain. For marketing, they provide a digital IQ Index allowing customers to assess their digital maturity online [7]. Even though it is specific for marketing, this model provides an interesting view of the nine dimensions taken into account as illustrated in the figure 2 below :

Figure 2: *Criteria of the Gartner Model*

4.2.1.2.3. Frost & Sullivan Model

The Frost and Sullivan model is slightly different from the previous ones as it aims to measure the digital transformation readiness of airlines [8]. They have developed indicators on a scale from one to five. Those indicators are themselves contributing to four different scoring parameters and the sum of these parameters gives a Digital Readiness Composite Score as shown in figure 3 below :

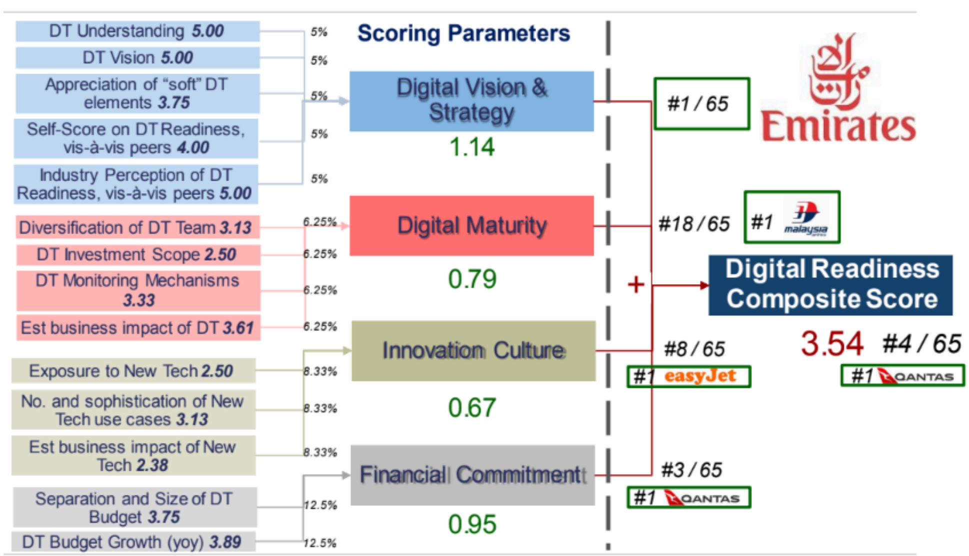

Figure 3: *Parameters of the Frost & Sullivan Model*

The only issue with this methodology is that it remains unclear what it measures. What does Digital Transformation Readiness mean? How can we derive insightful actions? There are high chances that the overall score doesn't mean anything and that one should rather look at the situation by domain then [4].

4.2.1.2.4. Lufthansa's Airline Digital Index Model

This model has been developed by the Lufthansa Innovation Hub. So far, this is the only model aiming at benchmarking the digital performance and maturity of airlines. It qualifies the degree of digitalisation of 26 major airlines worldwide through a total of 38 metrics (based on data collected from 2017 to 2019) aggregated into a final digital maturity score [9].

The interesting approach in the Airline Digital Index (ADIX) model is that it is composed of 100% quantitative criteria based on theoretically accessible data in order to be able to run the assessment and update it. It also details the criteria and the sources used.

The model is based on four main domains divided in categories, and for each category there may be several metrics. The dimensions considered in this model are :

- ➔ Digital Organisation
- ➔ Digital Strategy
- ➔ Digital Product
- ➔ Digital Engagement

4.2.2. Machine Learning for Customer Segmentation

Many literatures have reviewed the application of data mining techniques in customer segmentation, and achieved sound effectives.

While customer segmentation and market segmentation are often treated as interchangeable concepts in the literature, they differ significantly in terms of the data required for their clustering mechanisms. Market segmentation primarily focuses on acquiring new customers, which is the first stage of Customer Relationship Management (CRM), and relies on socio-demographic data. Customer segmentation, on the other hand, operates across all stages of CRM, utilising both socio-demographic and transactional data [10].

4.2.2.1. Statistical Methods

Clustering is a process of dividing a set of objects into groups of similar members making it very popular for customer segmentation. Classical clustering algorithms can be roughly divided into the following categories: partition-based methods, hierarchy-based methods, density-based methods, grid-based methods, and graph-based methods [11]. The table 2 below gives more insights on these categories.

Category	Definition	Popular Algorithms
Partition-based Methods	Divide the data into non-overlapping clusters, where each cluster is represented by a centroid or a prototype.	K-Means and variants (C-Means, K-Medoids, etc.)
Hierarchy-based Methods	Build a hierarchy of clusters by merging or splitting existing clusters, resulting in a tree-like structure.	Agglomerative Hierarchical Clustering, Divisive Hierarchical Clustering, BIRCH
Density-based Methods	Identify clusters as regions of high density in the data space, separated by regions of low density.	DBSCAN OPTICS DENCLUE
Graph-based Methods	Represent the data as a graph, where each data point is a node, and the edges represent the similarity between nodes.	Spectral Clustering Affinity Propagation
Grid-based methods	Divide the data space into a grid of cells and perform clustering on the cells. These methods are particularly useful for large datasets.	STING (Statistical Information Grid) CLIQUE (CLustering In QUEst)

Table 2:*Statistical Clustering Algorithms according to [11]*

Some of these models can even be combined with optimization algorithms to achieve better results. Yue Li et al. [12] use Adaptive Learning Particle Swarm Optimization (ALPSO), a variant of Particle Swarm Optimization (PSO), which is a population-based optimisation algorithm inspired by the social behaviour of bird flocking and fish schooling used to optimise the original K-means clustering algorithm by preventing its overdependence on the initial cluster centres.

These models are effective only when the features are representative, while their performance on the complex data is usually limited due to the poor power of feature learning. As such, Deep Learning methods have risen for their excellent nonlinear mapping capability and their flexibility in different scenarios [13].

4.2.2.2. Deep Learning Methods

Yazhou Ren et al. [13] classify deep learning methods from the perspective of initial settings of data and deep learning methodology, reducing it to four main categories as shown in figure 4 below. :

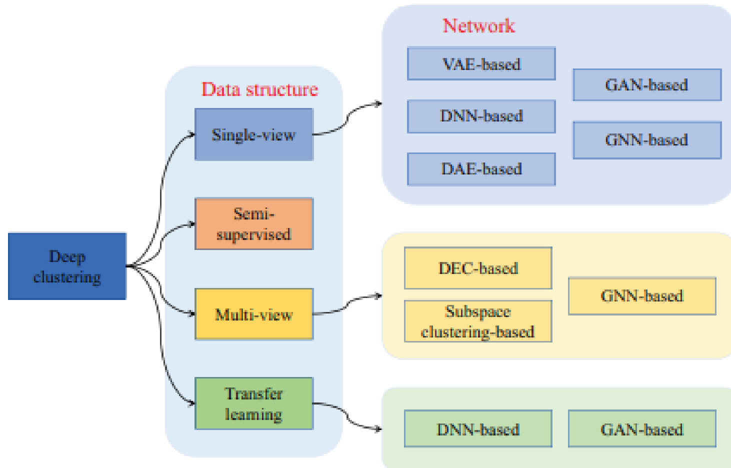

Figure 4*: Deep Clustering Algorithms according to [13]*

The first category of methods known as Deep single-view clustering use a single view or source of data to learn a clustering model. They do not rely on any additional information, such as labels or multiple views, to perform clustering using a single view or source of data to learn a clustering model.

Deep clustering based on semi-supervised learning methods use a small amount of labelled data in addition to a large amount of unlabeled data, to learn a clustering model. The labelled data provides some supervision to guide the clustering process through a combination of "must-link", "cannot-link" constraints. Samples with the "must-link" constraint belong to the same cluster, while samples with the "cannot-link" constraint belong to different clusters. These methods only deal with single-view data.

Deep clustering based on multi-view learning methods use multiple views or sources of data to learn a clustering model. Each view provides a different perspective on the data, and the clustering model is learned by integrating information from all views. For example, the report of the same topic can be expressed with different languages; the same dog can be captured from different angles by the cameras, etc.

Deep clustering based on transfer learning methods use pre-trained models or knowledge from one domain to learn a clustering model in another domain. The pre-trained models or knowledge provide a good starting point for the clustering model, which can then be fine-tuned for the target domain. The network structures used in most of these methods are autoencoders.
A set of examples for each category is given in [13].

4.3. Methodology

Two distinct use cases were explored. The first case was a general analysis, while the second case was more specific, concentrating on airlines that use Skywise. We opted for this dual approach due to the challenges we faced in collecting data for the general case, which led us to consider the more specific case, where internal data was readily available.

4.3.1. Data Sources & Dataset Creation

4.3.1.1. Broad Analysis

Based on the digital maturity models reviewed in the state of the art, our critical dimensions were selected to inform the construction of the dataset for the initial use case. Table 3 below gives more insights on the components of these dimensions:

Dimension	Data Collected	Data Sources
Fleet	Fleet Distribution (Airbus/Non-Airbus) Fleet Age	Cirium Skywise
Operations	Digital Solutions / Operations Provider(s)	Salesforce
Customers	Mobile Application Rating Scores Mobile Application Rating Counts	Apple AppStore via iTunes Search API
Organisation	IT Profiles IT Recruiting	Linkedin

***Table 3**: Data Sources for the Board Analysis*

Due to limitations in accessing the LinkedIn API, a PoC will be created using Skywise usage metrics for airlines already on the platform. This approach will provide access to a more comprehensive dataset, enabling the test and refinement of the algorithm. However, it is essential to note that hyperparameter optimization will be required once the dataset for the broad analysis becomes available to ensure optimal performance.

4.3.1.2. Skywise Users Analysis

The Skywise users focused analysis' goal is to provide a comprehensive understanding of how airline users engage with Skywise. Therefore, the input dataset of this use case will contain Skywise usage metrics for each airline for the past year, i.e. the number of unique accounts, the average number of connections per account and per day, and the average session length for each application available on Skywise, alongside fleet information (age and distribution).

4.3.2. Digital Maturity Assessment

4.3.2.1. Workflow

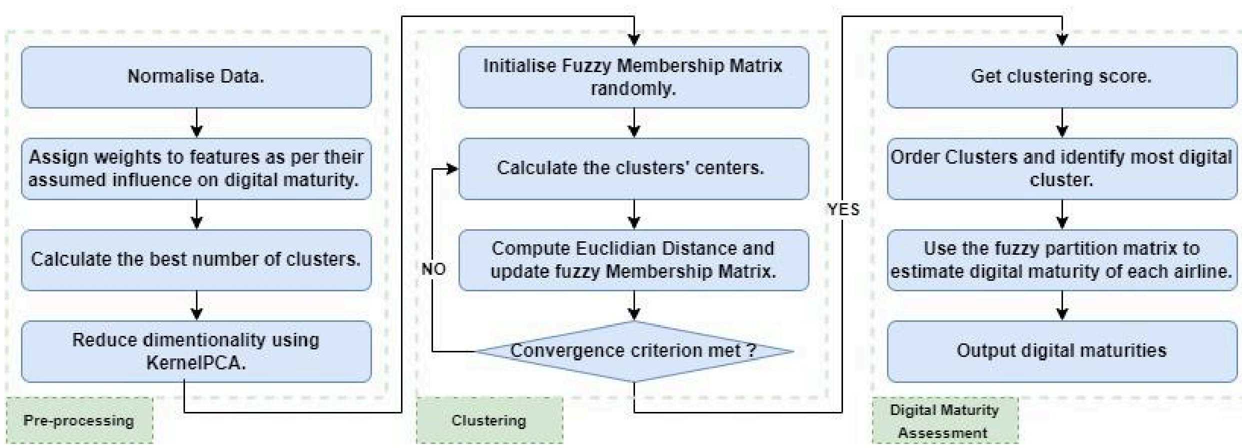

***Figure 5**: Workflow of the Digital Maturity Estimation Model*

The workflow initiates with preprocessing. To prevent feature dominance, normalisation is applied prior to clustering. To strike a balance between sample size and feature dimensionality, we employ KernelPCA for dimensionality reduction, which effectively handles nonlinear relationships between features. Subsequently, the optimal number of clusters is determined using the elbow method of k-means, ensuring a well-suited clustering configuration.

The clustering part will be conducted using Fuzzy C-Means (FCM) as it allows data points to belong to multiple clusters with varying degrees of membership. Unlike traditional k-means, which assigns each data point to a single cluster, FCM uses fuzzy logic to assign a membership value between 0 and 1 to each data point for each cluster. The algorithm iteratively updates the cluster centres and membership values to minimise the distance between data points and cluster centres. The membership values indicate the degree of similarity between a data point and a cluster, allowing for soft clustering and handling of noisy or overlapping data.

The resulting clusters and membership matrix will be used to rank airlines from most to least digital, and a coefficient assigned to each membership score to estimate their potential digital maturity. This approach was developed in collaboration with an experienced marketing manager who provided valuable insights into airline's behaviours. Furthermore, a feature importance analysis will be conducted to order the influence of input features on the predictions of the algorithm.

4.3.2.2. Pre-processing

The dataset of this analysis limited to the airlines already on Skywise contains only numerical features. They are scaled on the range [0, 1] using the the min-max scaling method with the following formula:

$$\overline{x_{ij}} = \frac{x_{ij} - min_i x_{ij}}{max_i x_{ij} - min_i x_{ij}} \tag{1}$$

This is an essential step in the data pre-processing phase for clustering algorithms as they are sensitive to feature scales. It will ensure that we master the contribution of each feature to the clustering algorithm. After scaling, a weight between 1 and 5 will be assigned to each feature of the dataset according to its estimated importance in the segmentation by marketing experts. This will allow the clustering algorithm to give more importance to features that are deemed more relevant for segmentation purposes.

To determine the optimal number of clusters for the dataset, we will employ the elbow method, a visual approach for identifying the point of diminishing returns in the within-cluster sum of squares (WCSS) as the number of clusters (k) increases. By plotting the WCSS against k, the 'elbow point', where the rate of decrease in WCSS slows down, indicates the most suitable number of clusters for our data. Figure 6 below shows the results for the dataset :

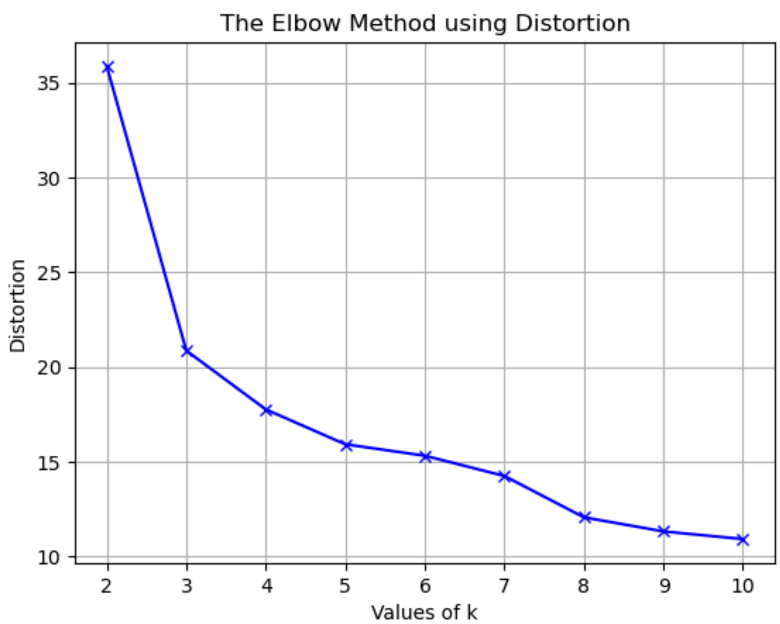

***Figure 6**: Elbow Method Plot*

The optimal number of clusters here is **3**.

Given the small size of the dataset (159 samples), dimensionality reduction is applied using Principal Component Analysis (PCA) to mitigate the risk of poor clustering performance. A typical rule of thumb when performing PCA is that the subset of the principal components should keep at least 80% of the original variance of the data [14]. The results of the dimensionality reduction experiments using both traditional PCA and Kernel PCA (KPCA) are presented in Table 4 below:

	PCA		Kernel PCA	
N° PCA Components	**Silhouette Score**	**Conserved Variance**	**Silhouette Score**	**Conserved Variance**
2	0.758	70.89 %	**0.804**	**86.12 %**
3	0.673	77.14 %	0.667	89.45 %
4	0.614	81.15 %	0.581	91.54 %
5	0.578	85.17 %	0.541	93.09 %

Table 4*: PCA Vs Kernel-PCA Comparison Metrics*

A trade-off between the clustering score (silhouette score) and the conserved variance was observed, leading to the selection of Kernel PCA with two components as the optimal approach.

A hyper-prarameters optimization for Kernel PCA was performed using Optuna but it won't be detailed here.

4.3.2.3. Clustering

The goal of this study was not merely to cluster the airlines, but to estimate a digital maturity score for each of them. To achieve this, we chose to use Fuzzy C-Means clustering (FCM). Unlike hard clustering algorithms, Fuzzy C-Means returns not only cluster labels but also membership values, which represent the degree to which each element belongs to each cluster.

The membership values will be used to estimate digital maturity afterwards, and propose the most suitable range of digital solutions for each airline.

Fuzzy c-means partitions the n samples of the dataset $X = \{x_1, x_2, ..., x_n\}$ in R^d dimensional space into c (1 < c < n) clusters with $Z = \{z_1, z_2, ..., z_c\}$ cluster centres or centroids. The fuzzy clustering is described by a membership matrix μ with c (number of clusters) columns and n (number of data points) rows where the element in the row i and column j in μ, indicates the degree of association or membership function of the i^{th} object with the j^{th} cluster [15]. The membership matrix μ is characterised by :

- $\mu_{ij} \in [0, 1]\ \forall i = 1, 2, ..., n; \forall j = 1, 2, ..., c$ (2)
- $\sum_{j=1}^{c} \mu_{ij} = 1\ \forall i = 1, 2, ..., n$ (3)
- $0 < \sum_{i=1}^{n} \mu_{ij} < n\ \forall j = 1, 2, ..., c$ (4)

The objective function of the fuzzy c-means algorithm is to minimise the following equation :

$$J = \sum_{j=1}^{c} \sum_{i=1}^{n} \mu_{ij}^{m} \left\| x_i - z_j \right\| \tag{5}$$

where m is the fuzziness exponent and z_j is the centroid of the cluster j that is obtained by the following equation :

$$z_j = \frac{\sum_{i=1}^{n} \mu_{ij}^{m} x_i}{\sum_{i=1}^{n} \mu_{ij}^{m}} \tag{6}$$

The FCM algorithm is iterative and can be stated as follows [15]:

```
Initialize the membership matrix values
While convergence criterion not met do
  | Compute the cluster centers
  | For each datapoint do
  |   | Compute Euclidian distance between the datapoint and the clusters' centers
  | End
  | Update the membership matrix
End
```

***Figure 7**: Fuzzy C-Means Algorithm*

The membership matrix will be initialised randomly and the convergence criterion is met if one of the following conditions is fulfilled:

- The maximum number of iterations t_{max} is reached
- Convergence of the membership matrix : $||\mu(t-1) - \mu(t)|| < \epsilon$ (7)

where ε is a small positive value (e.g., 1e-6) i.e. the minimum accepted error.

At each iteration, the membership matrix will be updated according to the equation below :

$$\mu_{ij} = \left[\sum_{k=1}^{c} \left(\frac{\left\| x_i - z_j \right\|}{\left\| x_i - z_k \right\|} \right)^{\frac{2}{m-1}} \right]^{-1} \tag{8}$$

4.3.2.4. Digital Maturity Estimation

The digital maturity estimation is based on the values membership matrix, which assigns each airline to a cluster. The assignment is determined by the cluster with the highest membership value for each airline. Subsequently, the clusters are ordered using a score, and the memberships are translated into digital maturity indices.

Considering $x_i = \{x_{i1}, x_{i2}, ..., x_{id}\}$ the i-th airline in the dataset, the score of each cluster is given by :

$$s_j = \frac{\sum_{x_i \in cluster_j} \sum_{k=1}^{d} \alpha_k \overline{x_{ik}}}{\sum_{x_i \in cluster_j} 1} \quad (9)$$

where :

- $\alpha_k = -1$ if $x_{ik} \in$ [“fleet_age”, airbus_feet_age”]
- $\alpha_k = 1$ if $x_{ik} \notin$ [“fleet_age”, airbus_feet_age”]
- $\overline{x_{ik}}$ the scaled value of x_{ik}

The choice of α_k is motivated by the fact that the fleet features ("fleet_age" and "airbus_fleet_age") are the only ones with a negative implication, i.e., a higher value indicates a less desirable outcome (older fleet). By setting $\alpha_k = -1$ for these features, we effectively penalise clusters with older fleets, while rewarding those with newer fleets. The cluster with the highest score is deemed more digitally mature.

With the ordering of clusters and assuming that cluster 1 is the most digitally mature followed by cluster 2 and so on till cluster c. The digital maturity index of each airline x_i will be calculated as:

$$\overline{DMI_i} = \sum_{j=1}^{c} (c - j)\mu_{ij} \quad (10)$$

To prevent a fragmented distribution of digital maturity scores, airlines will then be ranked by ascending digital maturity and the digital maturity indices will then be scaled up so that the highest digital maturity will be set to 95% and the minimum to 35% following the equation below:

$$DMI_i = (95 - 35) \times \frac{rank_i - 1}{max_i(rank)} + 35 \quad (11)$$

Where $rank_i$ is the rank of airline x_i by ascending digital maturity. This will ensure a smooth progression of scores. Without this approach, we risk having a disjointed distribution of scores, where one set of elements may cluster around a high score (e.g., 95), while another set clusters around a significantly lower score (e.g., 60).

4.4. Results

4.4.1. Clustering Results

The Fuzzy C-Means algorithm converged with a silhouette score of 0.804, which is a measure of how well each data point fits into its assigned cluster. The silhouette score ranges from -1 to 1, where a higher score indicates that the data point is well-matched to its cluster and a lower score indicates that it may be assigned to the wrong cluster. A rule of thumb for clustering is to aim for a silhouette score of 0.7 or higher, so our results are above this threshold, indicating a good clustering performance.

Below, a visualisation of the PCA representation of the data as well as its distribution among clusters.

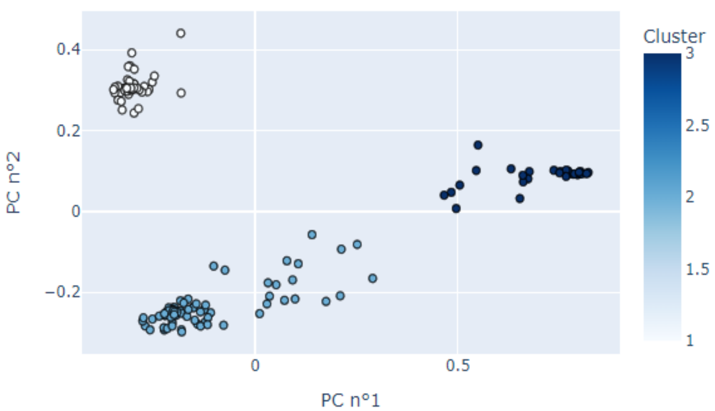

***Figure 8**: Clusters Visualisation*

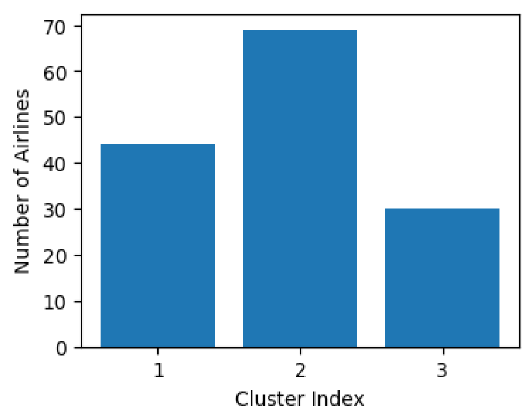

***Figure 9**: Clusters Distribution*

In figure 8, we can visually identify three distinct clusters confirming the algorithm results. Cluster 1 is more dense while clusters 2 and 3 are more “airy” explaining the silhouette score. Clusters were ordered as per digital maturity, meaning cluster 1 is the most digitally mature followed by cluster 2 and then cluster 3.

Figure 9 suggests that the algorithm is effective at identifying airlines at the extremes of digital maturity, but may struggle to accurately classify those in between. The feedback from the marketing experts supports this interpretation, as they were surprised by some airlines in Cluster 2 but not by those in Cluster 1 and Cluster 3. Given this, the algorithm's performance can still be considered acceptable, as it is able to accurately identify the extremes of digital maturity, which is consistent with expert knowledge.

Figure 10 below shows the digital maturity scores of airlines:

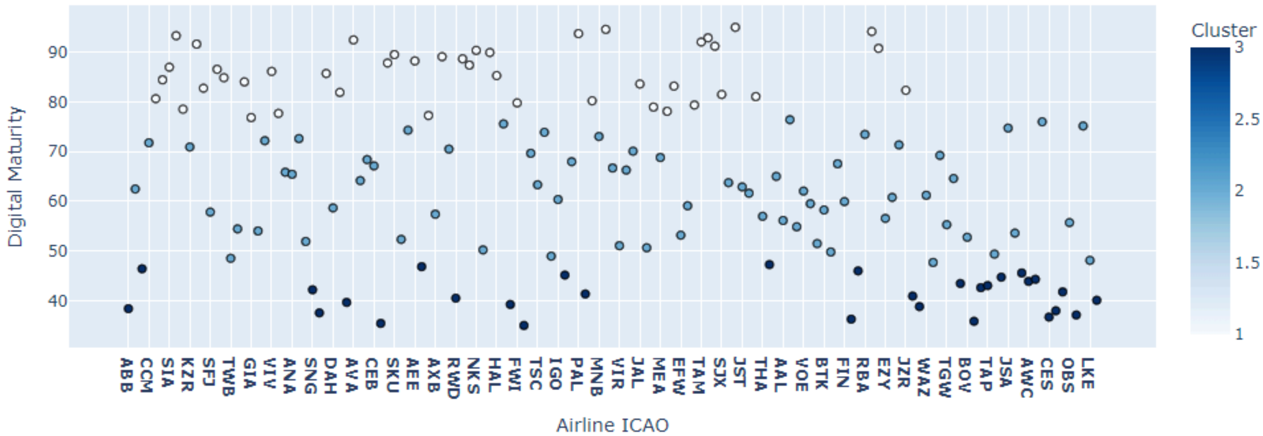

***Figure 10**: Digital Maturity Scores*

The top ten most digital airlines are summarised in table 5 below. They will be compared to the qualitative digital maturity assessment used before this study:

ICAO Code	Airline Name	Quantitative Digital Maturity	Qualitative Digital Maturity
AAY	Allegiant Air	95.00 %	Gold
EIN	Aer Lingus	94.58 %	Bronze
VLG	Vueling	94.15 %	Silver
PAL	Philippine Airlines	93.73 %	Below Bronze
QFA	Qantas	93.31 %	Gold
MAS	Malaysia Airlines	92.89 %	Gold
AVA	Avianca	92.46 %	Silver
TAM	LATAM Airlines Brasil	92.04 %	Gold
ETH	Ethiopian Airlines	91.62 %	Below Bronze
EXS	Jet2	91.20 %	Silver

***Table 5**: Top Ten Digitally Mature Airlines*

Four out of these airlines were classified as golden Skywise users in the qualitative classification while 3 were considered as silver users. However there are two supposedly bronze users and one below bronze.

Notably, Aer Lingus ranked among the top 10 digital airlines in the Airline Digital Index, published by Lufthansa, in multiple categories [8]. Specifically, they achieved:

- 2nd place in the share of workforce on LinkedIn with digital/IT-related job titles
- 8th place in digital strategy, outperforming both LATAM and Vueling
- A higher score than Vueling in Digital Engagement

This strong performance is reflected in their digital maturity score, which is influenced by the utilisation metrics for Skywise in our input dataset.

According to the 2024 Digital Maturity Report [16] published by Remarkable Group, Ethiopian Airlines has a digital maturity score slightly lower than that of LATAM matching the results or our algorithm. Philippine Airlines on the other hand is probably an outlier (misclassified) given that our silhouette score is not of 1.

A similar analysis for each airline in the dataset was performed; comparing the quantitative assessment with qualitative evaluations, expert opinions, and other assessments where possible.

4.4.2. Features Importance

Feature importances are calculated by permutation by randomly shuffling the values of a single feature and observing the resulting degradation of the silhouette score. Hence, it involves the importance of features for both dimensionality reduction and Fuzzy C-Means clustering. Figure 11 below shows both the calculated feature importances as well as the the initial assigned weight for each feature :

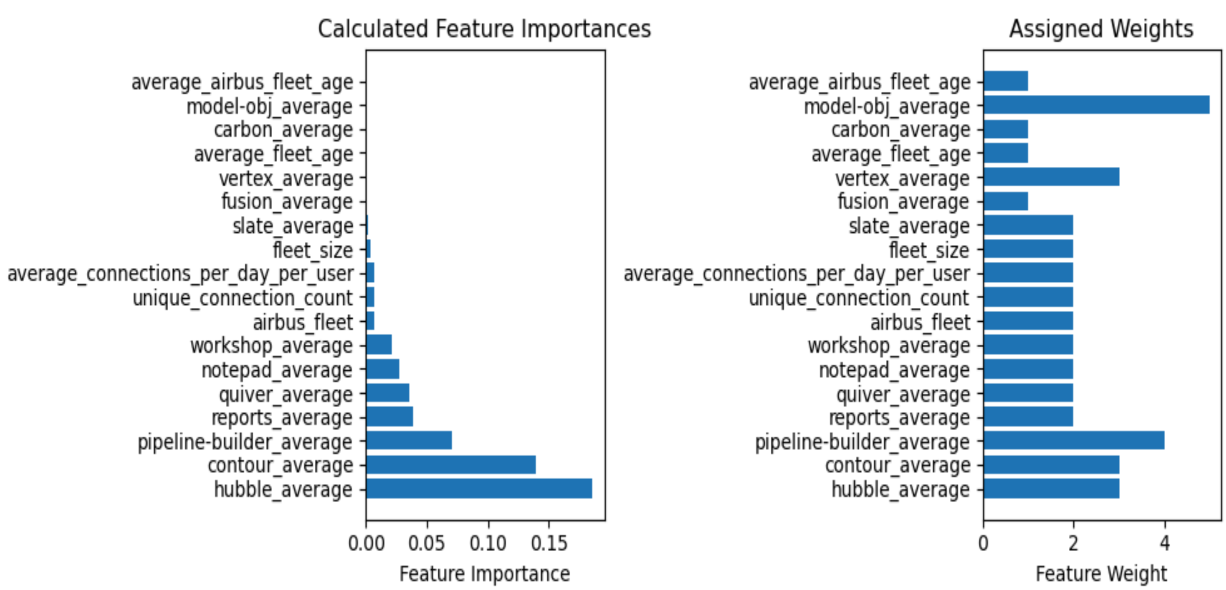

***Figure 11**: Feature Importances and Initial Weights for C-Means*

Aside from the model object app usage, the feature importances go along with the initially assigned weights. In fact the model object app (Modelling Objectives) is used to manage and deploy machine learning models and is not used by most airlines in the dataset as shown in figure 12 across. Modelling Objectives being an indicator of a higher digital maturity, we will keep it in the dataset anticipating future sales.

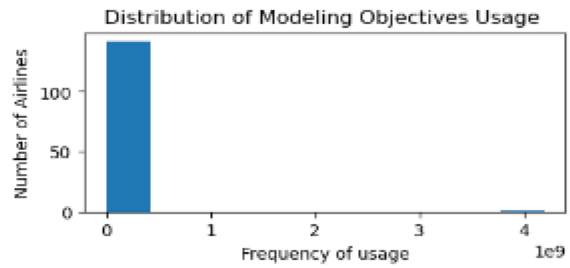

***Figure 12**: modelling objective usage*

4.5. Conclusion and Future Work

The primary goal of this study is to develop a digital maturity index for each airline. As there was no prior estimation of digital maturity, we explored unsupervised learning approaches to achieve this objective. Specifically, fuzzy clustering using C-Means is employed to group airlines and the resulting membership matrix is combined with expert-defined rules to create a hybrid estimation of digital maturity.

To validate the approach, the results are compared with expert knowledge and other digital maturity estimations. However, due to limitations in data availability, only the digital maturity scores of the most digital airlines are considered for comparison. This allowed us to assess the effectiveness of our approach in identifying top performers.

A dashboard has been developed in Skywise to visualise the outcomes of this study. The dashboard comprises two tabs: the “Airline Profiles” tab, which presents a comprehensive overview of the findings, including a plot of digital maturities, cluster segments, and insightful observations on how each airline utilises the platform; and the “Methodology & Evaluation” tab, which provides a concise summary of the model's functionality, key performance metrics for monitoring its performance (the silhouette score, conserved variance, and date of last update), as well as feature importances. Screenshots of both tabs are included in Appendices A and B, respectively.

The study on Skywise users was conducted as a proof-of-concept (PoC) for the broader analysis. This decision was made due to difficulties in obtaining certain components of the broader analysis dataset. Although the two datasets do not share identical features, the algorithm is designed to be feature-agnostic, allowing it to be applied to the broader dataset with minimal modifications. Nevertheless, a new hyperparameter optimization will be necessary to ensure optimal performance. In the meantime, the dashboard will be used by the marketing department to follow-up on the evolution of Skywise usage for existing customers (and future ones). Since Skywise proposes several chargeable tiers, the marketing department will use the dashboard to better target future upsale opportunities, as they will be closely related to the evolving digital maturity of airlines.

Building on the experience gained from this market segmentation analysis, a second study is performed to cover the duration of the apprenticeship. This second sub-study aimed to forecast equipment degradation using the operating conditions of the aircraft, with the ultimate goal of reducing downtime, and optimising maintenance schedules. The next sections will discuss in further details the context and implementation of this solution.

5. Predictive Maintenance

5.1. Objectives

Aircraft maintenance is not only essential for ensuring safety, but also for maintaining a high level of reliability and service quality that meets passenger expectations. Even when an aircraft is still airworthy, maintenance can be performed proactively to prevent potential issues and minimise operational interruptions. This proactive approach is critical in the aviation industry, where unplanned downtime can lead to flight delays, cancellations, and reputational damage.

The objective of this second study is to develop a predictive model that forecasts the degradation index of a non-"No-Go" equipment based on the environmental conditions in which the aircraft has operated. Predicting the degradation index of this equipment shall allow to:

- ➔ Monitor the equipment's health, allowing for better forecasting of the next maintenance slot.
- ➔ Understand the reasons behind degradation, allowing to correct this issue for aircrafts to be delivered in the future.

To ensure confidentiality, the equipment and dataset used in this study will be anonymized.

5.2. State of the Art

5.2.1. Definitions

Maintenance is defined by the norm EN 13306 [17] as the combination of all technical, administrative and managerial actions during the life cycle of an item intended to retain it in, or restore it to, a state in which it can perform the required function.

Maintenance strategies can be classified into three types depending on their trigger, curative maintenance, preventive maintenance and predictive maintenance [18].

Curative or corrective maintenance intends to eliminate the cause and effect of a detected failure or functional loss on a component in order to immediately restore its operational status. From Airbus's perspective, curative maintenance is synonymous with “Run to failure”, it can therefore be unplanned maintenance (further to unexpected event) or planned maintenance

(Non Routine task further to findings during scheduled maintenance: for instance scheduled maintenance to cover hidden failures) [19].

Preventive maintenance includes maintenance tasks performed on a component in working condition and intends to assess and/or mitigate its degradation to prevent its failure or fault and associated curative maintenance [19].

Predictive maintenance intends to assess and analyse the physical conditions of a component before scheduling and triggering the required maintenance action at the accurate moment [17]. Predictive maintenance is not a new topic as it already existed in the 1940's [20]. However, it has been getting more attention in the last couple of years due to its benefits in terms of reliability, safety and maintenance cost [20]. Predictive maintenance has mainly three objectives [18]:

- ➔ Reduce unplanned maintenance to a minimum.
- ➔ Eliminate or optimise planned maintenance.
- ➔ Optimise efficiency of remaining maintenance actions (durations and costs).

In literature, predictive maintenance is often associated with prognostics, health monitoring (PHM) and condition-based maintenance (CBM). While PHM and CBM are sometimes introduced as synonyms to predictive maintenance, they are also considered as extensions of it [20]. From Airbus's perspective, a key distinction is made: if a remaining useful life (RUL) is computed, it is referred to as condition-based maintenance; otherwise, it is considered predictive maintenance [19].

Jimenez et al.'s publication [20] categorises predictive maintenance approaches into single-model and multi-model types. The same taxonomy will be adopted for this bibliographic review and the following sections 5.2.2 and 5.2.3 give more insights on both types of approaches and highlight the sub-methods for each of them.

5.2.2. Single-model approaches for predictive maintenance

Following the taxonomy presented in [20], single-model approaches are either knowledge-based, data-driven, or physics-based.

5.2.2.1. Knowledge-Based Models

A knowledge-based predictive maintenance system (KBS) is designed to mimic the decision-making process of a human expert using a knowledge base that contains domain-specific facts, rules, and experiences [20, 21]. These approaches have been applied in industry since the beginning of the 1990s [20, 21].

Some examples of knowledge-based models in predictive maintenance include:

- ➔ Rule-based systems: These systems code human reasoning in the form of rules "IF-THEN" to reason about the system's behaviour. If these rules assume continuous values according to their degree of truthfulness, the rule-based systems will be referred to as fuzzy knowledge-based systems.

➔ Case-based systems: These systems use historical cases and experiences to reason about the system's behaviour.

While knowledge-based models can provide transparent and explainable decision-making processes and results, they are limited by expert knowledge that is either expensive or not available for all components or machines [20]. It is also important to note that these models may be used together with data-driven approaches, which will be discussed further in the next section [21].

5.2.2.2. Data-Driven Models

Data-driven approaches exploit information collected by sensors and actuators, in order to extract meaningful knowledge from it [21]. Jimenez et al. [20] classifies it into three groups detailed in the following sections.

5.2.2.2.1. Statistical Models

The estimations of these models are constructed by comparing the behaviour of the system of interest against known historical behaviours without relying on any physics or engineering principles [22]. The most popular statistical approaches include regression, autoregressive models, and Bayesian models and they are often used in combination with other models [20].

While statistical models offer many solutions for predictive maintenance, they require sufficient data to manage uncertainty [20] .

5.2.2.2.2. Stochastic Models

The main stochastic models to be found in literature are Wiener process, Gamma process, Gaussian process and inverse Gaussian process models [22].

In a Wiener process (also known as a Brownian motion), the degradation is modelled as a random walk with a drift term, which represents the deterministic component of the degradation, and a diffusion term, which represents the random fluctuations in the degradation process. One of the distinct features of the Wiener process is that it is a non-monotone stochastic process.

In a Gamma process, the degradation is modelled as a cumulative process, where the system's state at any given time is the sum of all the previous degradation increments. The Gamma process is often used to model the degradation of systems that exhibit monotonic and gradual damages such as fatigue, wear, corrosion, erosion etc. and they are only effective in such cases.

Both the Wiener process and the Gamma process are types of Markov processes which state that the future state of the process depends only on its current state, and not on any of its past states. They are therefore constrained by the assumption of Markov property.

A Gaussian process is a distribution over functions, where every point in the input space is associated with a Gaussian distribution. This probabilistic nature of Gaussian processes makes it challenging to apply the classic optimization approach; a probabilistic approach to optimization like the maximum likelihood method is more suitable.

The Inverse Gaussian process is a stochastic process that is defined as the mathematical inverse of a Gaussian process. It is a monotonic process, meaning that its values are either always increasing or always decreasing.

5.2.2.2.3. Machine Learning Models

Currently traditional Machine Learning models outperform statistical models when sufficient historical data exists. They both are outperformed by deep learning models when dealing with complex patterns and relationships in large datasets. Machine and Deep Learning models need two additional steps compared to the aforementioned models. These steps are data preprocessing and feature engineering, and are essential to enhance the accuracy of the models [23].

The table 6 below summarises some of the most popular machine and deep learning models used for predictive maintenance:

Model	Reference	Learning Process	Case Study
Extreme Gradient Boosting (XGBoost)	[24]	Supervised ML	Production lines in manufacturing
Random Forest	[25]	Supervised ML	Ship electric propulsion system
Self-Organizing Map	[26]	Unsupervised DL	Aircraft jet engines
Support Vector Machine (SVM)	[27]	Unsupervised DL	Fault detection in chemical industry
Convolutional Neural Networks (CNN)	[28]	Supervised DL	Bearing diagnosis
Recurrent Neural Networks (RNN)	[29]	Supervised DL	Jet engines

***Table 6**: Machine & Deep Learning Algorithms for Predictive Maintenance*

5.2.2.3. Physics-Based Models

Physics-based models, also known as model-based systems, use a physical understanding of the system to evaluate the remaining useful life (RUL) of an equipment [20].

These approaches, like the knowledge-based models, are domain specific and require deep knowledge on mathematics together with expertise on the physical behaviour of machinery's elements, which is expensive, time consuming, and often scarce for the majority of the components [21].

5.2.2.3. Multi-model approaches for predictive maintenance

Multi-model approaches, as the name suggests, are methods that use two or more of the previously cited methods. They are increasingly common in predictive maintenance tasks, especially when dealing with complex systems and multiple failure modes as they can overcome the limitations of single-model approaches [20]. Models in multi-model approaches can be configured sequentially, in parallel or one fully embedded into the other as shown in figure 13 below:

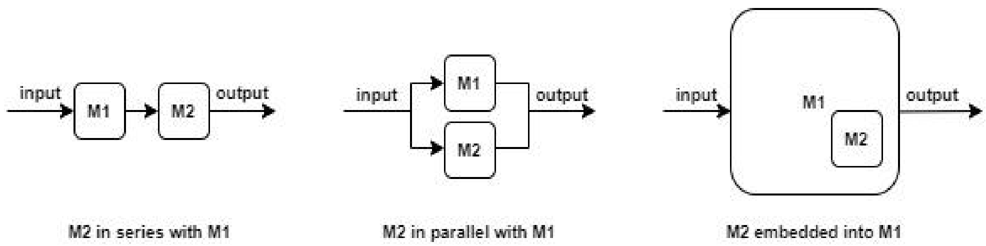

***Figure 13**: Configurations of Multi-Model Approaches*

The diversity of single-model approaches for predictive maintenance gives rise to a multitude of potential multi-model combinations. Table 7 provides an illustrative example of each possible multi-model configuration :

Combination	Reference	Case Study
Multiples knowledge-based models	[30]	Problem diagnostic in IT services
Multiple data-driven models	[31]	Remaining useful life estimation for IOT based systems
Multiple physics-based models	[32]	Diagnosis and prognosis of cracked rotor shafts
Knowledge-based models with data-driven models	[33]	Prognostics in aero-engines
Knowledge-based models with physics-based models	[34]	Prognostics tasks on mechanical parts
Data-driven models with physics-based models	[35]	Failures of rubber-metal-elements in technical systems

***Table 7**:Combinations of Multi-Model Approaches*

5.3. Methodology

5.3.1. Overview

The goal of this study extends beyond mere prediction of the degradation index, instead seeking to reveal the underlying mechanisms and factors contributing to its occurrence. Self-Organizing Maps (SOMs), a type of unsupervised neural network (UNN), are particularly well-suited for this task, as they are commonly used for clustering. By grouping similar data points together, it facilitates a deeper understanding of the degradation index's behaviour and the factors influencing its occurrence.

The proposed method leverages the strengths of SOMs by employing a two-phase approach, where a traditional unsupervised SOM clustering phase is followed by a supervised estimation phase to accurately predict the degradation index.

5.3.2. Self-Organizing Maps

Kohonen introduced self-organizing maps (SOM) for the first time in the 1980s [36]. They have been applied mainly in dimensionality reduction, data visualisation and unsupervised clustering as input to other models.

Self-organizing maps differ from other artificial neural networks as they apply competitive learning (each data point in the data set recognizes themselves by competing for representation) as opposed to error-correction learning (such as backpropagation with gradient descent), and in the sense that they use a neighbourhood function to preserve the topological properties of the input space.

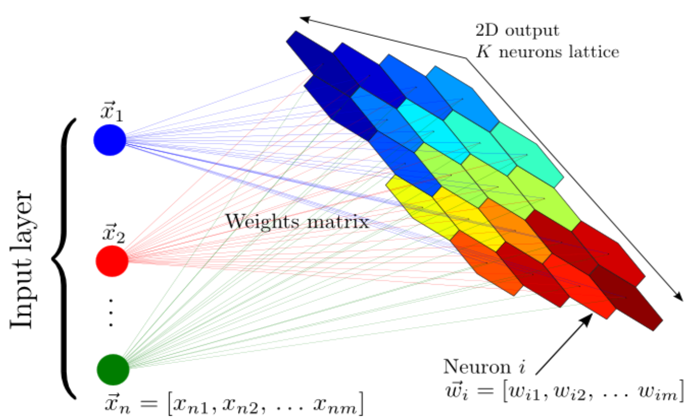

***Figure 14**: Self-Organizing Map*

While the first created SOM is a 2D hexagonal map, its topologies today can be in one, two or even three dimensions [26]. For this study, the easiest topology i.e. a 2D square map is used so that we can visualise it as a picture.

5.3.3. Input Data Management

To maintain confidentiality, all dataset details, and equipment specifications will be anonymised throughout this report, while maintaining the integrity and validity of the methodology and results presented.

The dataset contains 27 features and 579 samples, with only 49 samples that are labelled. The target variable, namely the degradation index is a continuous value ranging from 0 to 1. A critical threshold of 0.7 has been established, beyond which the maintenance task is triggered, indicating the need for intervention.

The input dataset is normalised on the range [0, 1] using the min-max scaling method with the formula (1) defined above. Ten of the labelled data points will be used in the test set and the rest of the data point will constitute the training set.

5.3.4. Parameters of the Map

According to [26], the size of the SOM can be determined from the number of observations or samples in the dataset such as :

$$M \simeq 5\sqrt{N} \tag{12}$$

Where M is the number of neurons and N is the number of samples in the dataset. For a square map we will have the number of rows/ columns as:

$$n = \lfloor\sqrt{M}\rfloor \tag{13}$$

The number of iterations for SOM convergence is then calculated following the rule of thumb cited by [26] as follows :

$$N_{iterations} = 500\, n^2 \tag{14}$$

5.3.5. Unsupervised Learning Phase

The unsupervised phase of the Self-Organizing Map (SOM) training algorithm is characterised by an iterative process, which is illustrated in Figure 15 below [26]:

```
Initialize the SOM weights
For t from 1 to Number_Of_Iterations do
  | Pick a random input vector among the input dataset
  | For each node of the map do
  |   | Compute Euclidian distance between the input vector and the map node weight vector
  |   | Find the best matching unit (node with the smallest distance)
  | End
  | Update the weight vector of the neighbourhood of the best matching unit and itself
End
```

Figure 15*: Unsupervised Self-Organizing Map Algorithm*

The weights of the neurons (nodes) will be uninitialised using a uniform distribution in the range of [0, 1] with a probability density function of 1 following the work of Schwartz, S. et al. [26]

The weight vector is updated at each iteration as follows :

- $w_{ij}(t + 1) = w_{ij}(t) + h_{kl,\,ij}(t)\,[x(t) - w_{ij}(t)] \quad \forall n_{ij} \in \Gamma_{BMU}$ (15)
- $w_{ij}(t + 1) = w_{ij}(t) \quad \forall n_{ij} \notin \Gamma_{BMU}$ (16)

Where:

- w_{ij} the weight vector of the node of coordinates (i, j)
- t the t^{th} iteration
- $h_{kl,\,ij}$ the neighbourhood function
- $x(t)$ the observed input vector
- kl the coordinates of the BMU (Best Matching Unit node)
- Γ_{BMU} the space of the neighbourhood of the BMU node and itself

Γ_{BMU} is defined by the width of the neighbourhood function, also called the BMU radius. Mtsushita et Nishio [37] proposed a Gaussian neighbourhood function defined by :

$$h_{kl,\,ij}(t) = \alpha(t)\, exp\,(- \frac{||w_{kl}(t) - w_{ij}(t)||^2}{2\sigma(t)^2}) \quad (17)$$

Where $\alpha(t)$ and $\sigma(t)$ are, respectively, the learning rate and the radius or width of neighbourhood function. Both $\alpha(t)$ and $\sigma(t)$ decrease over time for a faster convergence and to prevent oscillations following the equations below:

- $\alpha(t) = \alpha_0\,(1 - t/t_{max})$ (18)
- $\sigma(t) = \sigma_0\,(1 - t/t_{max})$ (19)

The initial learning rate α_0 is set to 0.9 and the learning rate function is limited to a minimum of 0.01 [36].

The initial radius σ_0 is set to half of the map size [36] and the radius function is limited to a minimum of 1 [38].

The BMU node n_{kl} is defined by:

$$n_{kl}\,|\,x(t) - w_{kl}(t) = arg_{ij}\,min\,[x(t) - w_{ij}(t)] \quad (20)$$

A node is considered seen if its weights vector has been updated at least once. Upon reaching the maximum number of iterations, any remaining unseen nodes will have their weight vectors computed as the average of their seen adjacent nodes (nodes that touch it in all four directions: up, down, left and right). This will be done in a row-major order, iterating over the nodes in ascending order of their (i, j) coordinates.

Once the maximum number of iterations t_{max} is reached and no node remains unseen. The unsupervised SOM is fully trained. It is now considered capable of finding the BMU of each and every datapoint regardless if it belongs to the training set or not. The Calculation of the BMU in the supervised training will be based on the trained unsupervised SOM.

5.3.6. Supervised Learning Phase

Following the framework of Riese, F. et al [38], A second SOM is attached to the fully trained unsupervised one. Both SOMs are of the same size.

The weight vectors of the unsupervised SOM are of the same dimension as the input data points while the weight vectors of the supervised SOM have the same dimension of target variable (1 in this study).

The workflow of the supervised training phase is similar to the unsupervised phase, except that the best matching units (BMU) are calculated using the unsupervised SOM. Figure 16 below shows the steps of the supervised training phase :

```
Initialize the supervised SOM weights
For t from 1 to Number_Of_Iterations do
  | Pick a random input vector among the input dataset
  | For each node of the map do
  |   | Find BMU based on the trained unsupervised SOM
  | End
  | Update the weight matrix of supervised SOM
```

Figure 16*: Supervised Self-Organizing Map Algorithm*

Note that the weights of the supervised SOM will be updated using the target variable while the weights of the unsupervised SOM were updated using the features.

The weight matrix of the supervised SOM will be uninitialised randomly, and the weights of the unseen nodes will be completed upon reaching the maximum number of iterations following the same logic as in the unsupervised training phase.

The output prediction process will involve the following steps:

1. The unsupervised SOM will be employed to determine the BMU for each data point
2. The weight vector of the corresponding BMU node in the supervised SOM will be output as the predicted value.

5.4. Results

Given that only 8.47 % (49 out of 579) of the dataset was labelled, two distinct runs were performed: one with full supervision on the labelled train data and another with semi-supervision on the entire train dataset.

5.4.2. Fully Supervised SOM

5.4.2.1. Experimental Setup

In this case the training set has 39 (49 -10) samples resulting according to equations (12) and (13) in a 6×6 map and 18000 iterations for each phase of the training.

This first run, leads to a root mean squared error (RMSE) of 0.21 on the test set. Unfortunately, this result is higher than our desired threshold of 0.1, indicating that the model's performance is not satisfactory with the available labelled data. This is because a map size of 6×6 generates 36 possible outputs which is very limited discretisation given that the target is a continuous value between 0 and 1. Figure 17 below shows the evolution of the RMSE in test depending on the size of the SOM :

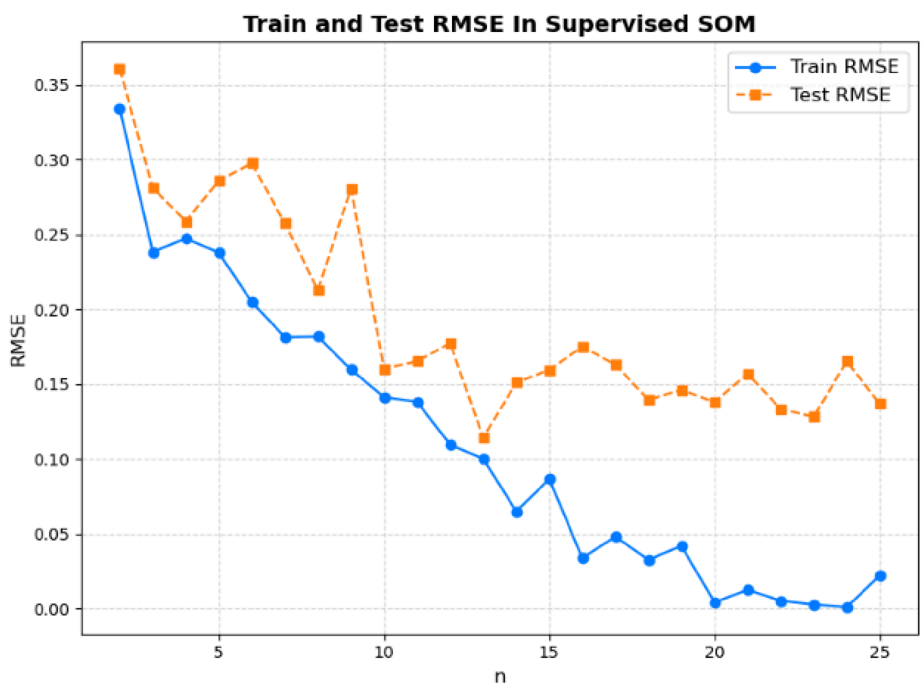

***Figure 17**: Evolution of the RMSE with the SOM Size In Supervised Run*

As illustrated, both the training and test root mean squared errors (RMSE) decrease as the Self-Organizing Map (SOM) size increases. This is likely due to the increased number of possible outcomes. A minimum RMSE is reached when the SOM size is 13×13 after which the difference between the training and test RMSE begins to diverge, indicating a sign of overfitting. This may be attributed to the fact that larger SOMs require a higher number of iterations, which, in the absence of early stopping, can lead to over-learning of the training dataset, resulting in overfitting.

Table 8 below summarises model parameters and performance metrics used in this fully supervised run:

Model Hyperparameters			Model Performance Metrics			
N° rows / columns	N° unsupervised iterations	N° supervised iterations	Train MAE	Train RMSE	Test MAE	Test RMSE
13	84500	84500	0.058	0.100	0.102	0.111

***Table 8**: Supervised SOM Parameters & Performance Metrics*

5.4.2.2. SOM Visualisation

While the performance metrics provide a quantitative assessment of the SOM's effectiveness, they alone are insufficient to fully understand the model's behaviour. Figures 18 and 19 below presents the SOMs generated respectively via the unsupervised and supervised training :

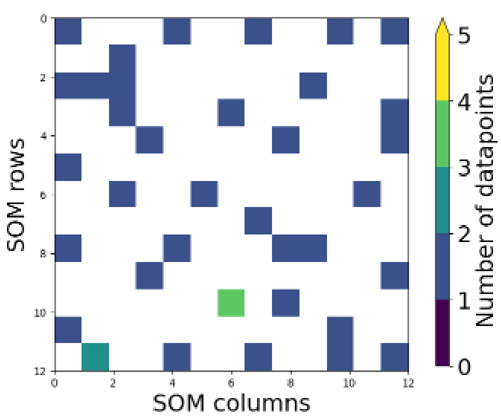

***Figure 18**: SOM After Unsupervised Training*

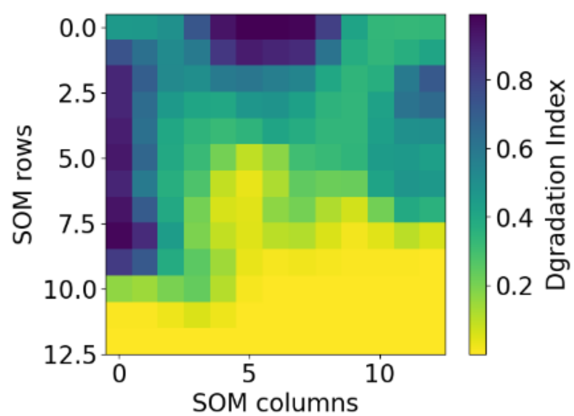

***Figure 19**: SOM After Supervised Training*

As shown in figure 18, each node is assigned to three or four data points at maximum. This is rather a good sign; as it indicates that the SOM is large enough to accommodate all data points and that no dominant cluster is found in the data suggesting no bias towards a particular subset of samples.

On the other hand, there are a lot of nodes with no data points assigned to them. While this is not unexpected, given that the number of nodes in the SOM exceeds the number of samples in the dataset of this run , it does have implications for the prediction. Specifically, the weights of these unassigned nodes are computed based on their neighbours, without being directly influenced by any samples from the dataset. This may explain the loss of precision translated via the relatively high RMSE of the model.

Figure 19 (to the right) is a representation of the predicted degradation index depending on the found Best Matching Unit (BMU). The possible outcomes range from 0 to 1 covering the entire range of the target variable. While nodes with a lower outcome are grouped together, there are three distinct clusters for nodes with higher degradation index. This is rather interesting as it suggests the presence of multiple root causes of degradation. More in-depth analysis of these clusters is conducted in the following section.

5.4.2.3. Cluster Analysis

To better understand what's driving the high degradation indices, we'll take a closer look at the points that make up each cluster and analyse their feature importances separately. The three clusters in question will be referred to as A, B, and C, as shown in figure 20 below:

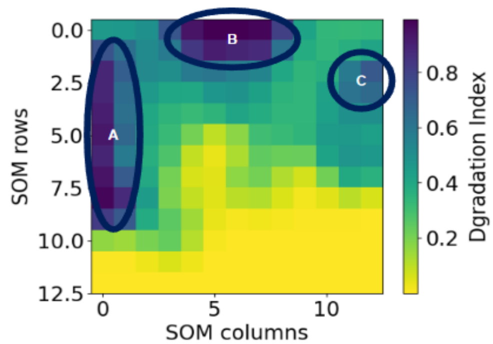

***Figure 20**: SOM Clusters in Supervised Run*

Figure 21 below shows the contribution of each feature in the model predictions for cluster A:

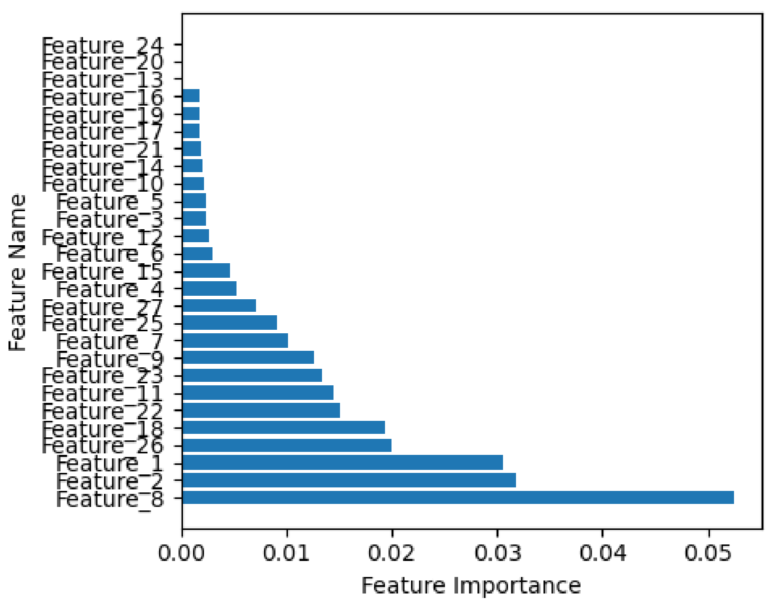

***Figure 21**: Feature Importances for Cluster A*

Examining the feature importance plot for cluster A, features 8, 1, and 2 are the main drivers of the model's decisions, followed by features 26, 18, and 23. The remaining features have a negligible contribution. This is interesting, because features 1 and 2 that are driving the model's decision are related to the manufacturing of the component and not to the way and condition of operating the aircraft.

The values of features 1 and 2 for the samples of cluster A are over the mean value of the dataset. This suggests that an increase in those values is likely to lead to an augmentation of the degradation index, leading to a sensitivity to the environmental factors, as predicted by the engineers and could indicate a quality control issue or a problem with the manufacturing process that needs to be addressed.

The features importance for Cluster B is shown in figure 22 below:

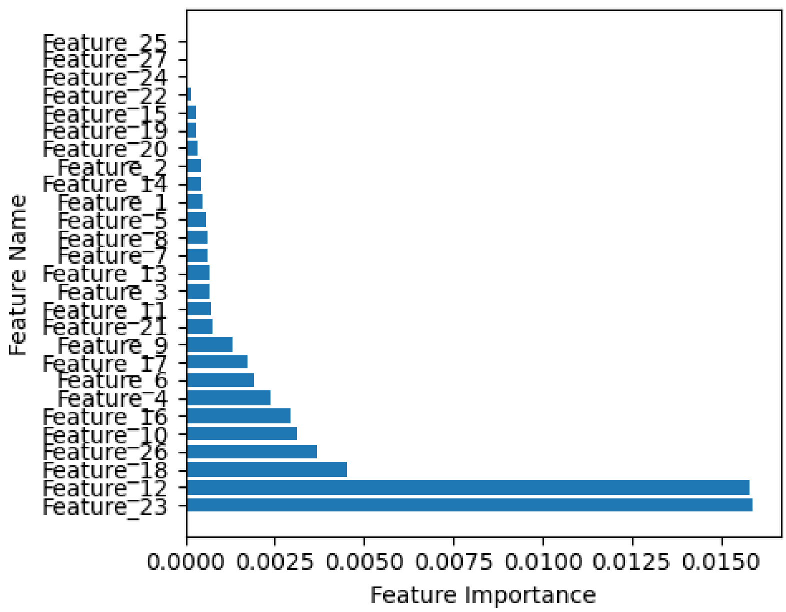

***Figure 22**: Feature Importances for Cluster B*

Cluster B, reveals a different finding, because features 12 and 23 that are mainly driving the model's decision are related to the exposure of the aircraft to certain environmental conditions. The values of these features for the aircrafts in this cluster are way over the mean value of the dataset reflecting an excessive exposure to these environmental conditions.

The aircrafts of this cluster are among those with highest flight cycles, meaning that they have been operated enough to explain this exposure. Features 12 and 23 then can be considered as the main causes of degradation of aircrafts in cluster B which is consistent with the regions in which these aircrafts are being operated.

Figure 23 below shows the feature effects for cluster C:

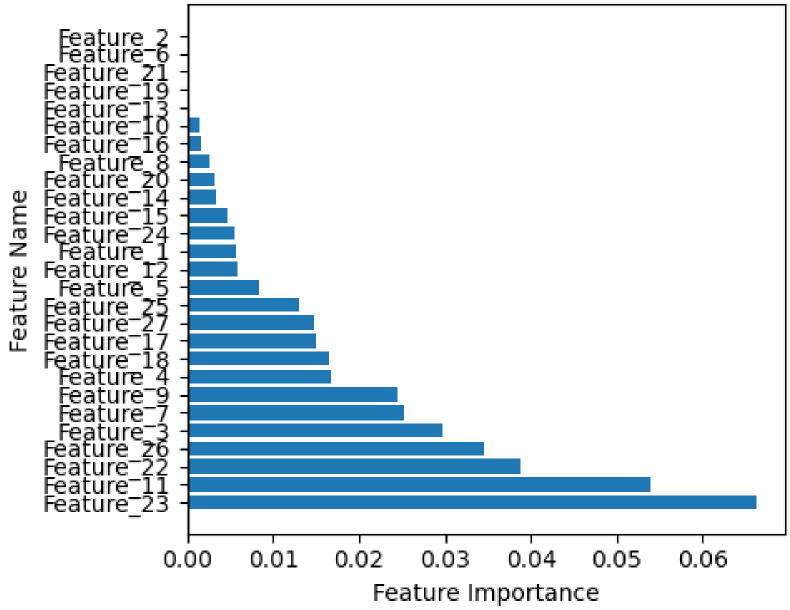

***Figure 23**: Feature Importances for Cluster C*

Features 23, 11, and 22 are the main drivers of the model followed by features 3, 26, 7, 9, and 4 and the rest of the features are way less important. Future 11 is the same as feature 22, except that feature 22 is measured on ground while feature 11 is measured in flight. Thus the feature importances makes sense since aircrafts of cluster C have higher flight hours than aircrafts of cluster B.

By combining the results from Clusters B and C, we can conclude that excessive exposure to the conditions measured by features 23, 11 (or 22) enhances degradation, regardless of whether the exposure occurs on the ground or in flight. The degradation for aircraft in Cluster B is higher than that for aircraft in cluster C, which may indicate that exposure on the ground is more detrimental.

5.4.3. Semi-Supervised SOM

5.4.3.1. Experimental Setup

In this second run, even the samples for which no degradation index is measured will be used in the unsupervised SOM to estimate BMUs. Since SOMs are not compatible with NaN values, these will be removed, and the same test set as the previous run will be retained. The training set will hence contain 546 samples resulting according to equations (12) and (13) in a 11×11 map and 60500 iterations for each phase of the training. However, informed by the insights gained from the first run, various map sizes will be explored to optimise performance.

Figure 24 below illustrates the evolution of the Root Mean Squared Errors (RMSE) on the training and test sets as a function of the SOM size:

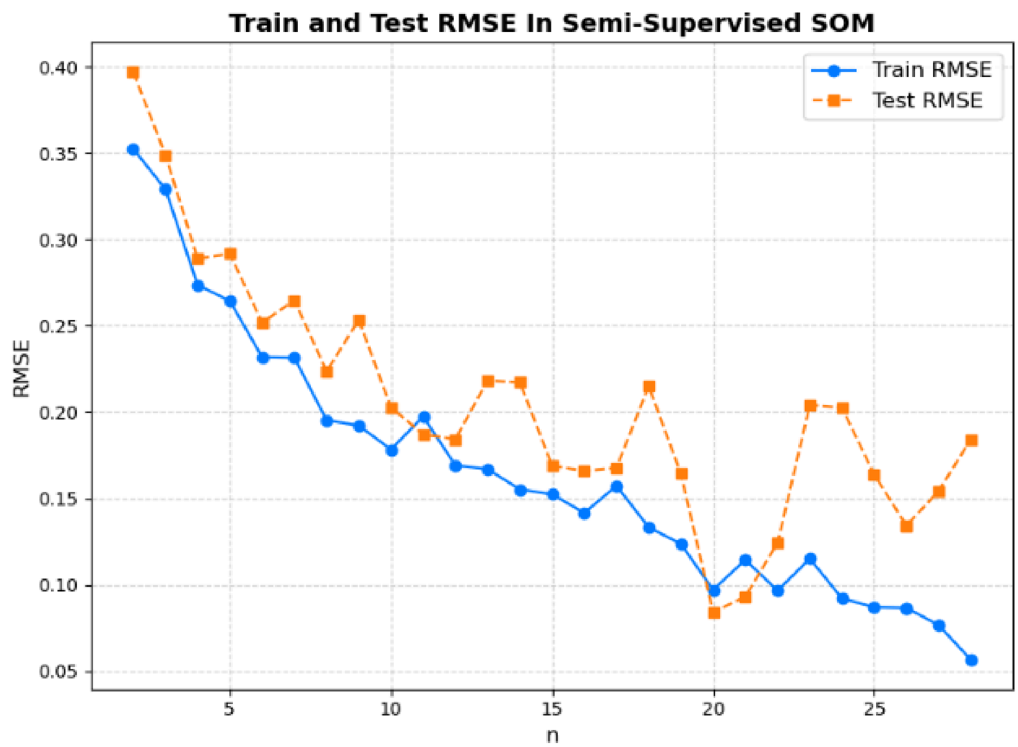

***Figure 24**: Evolution of the RMSE with the SOM Size In Semi-Supervised Run*

Notably, the results reveal that for a map size of 11×11 the RMSE is lower in test than it is in train which could indicate that the model is generalising really good in this case. However, it is not the best performance that we could get. This could be due to the fact that SOMs are mainly used for clustering, where multiple data points are expected to share the same Best Matching Unit (BMU). In our study, however, each BMU is assigned a predicted value, resulting in all points with the same BMU being assigned the same predicted value, which can lead to a loss of model accuracy. Therefore, it is reasonable to expect that larger maps would yield better results, until the maps become too large to be covered by the data points.

Compared to the one in the fully surprised learning run, in this convergence plot, the difference between the train and test RMSE are lower. This indicates that adding more data points to the unsupervised SOM allows it to capture data patterns more effectively and to estimate the BMU more accurately which is what is the goal of the semi-supervised run.

The optimal performance is achieved with a SOM of size 20×20. Similar to the 11×11 configuration, this SOM also exhibits a lower RMSE on the test set compared to the training set, instilling confidence in its generalisation capabilities.

Table 9 below provides a summary of the model parameters and performance metrics employed in this semi-supervised run:

Model Hyperprameters			Model Performance Metrics			
N° rows / columns	N° unsupervised iterations	N° supervised iterations	Train MAE	Train RMSE	Test MAE	Test RMSE
20	200000	200000	0.050	0.097	0.061	0.084

***Table 9**: Semi-Supervised SOM Parameters & Performance Metrics*

5.4.3.2. SOM Visualisation

Figures 25 and 26 below presents the SOMs generated respectively via the unsupervised and supervised training :

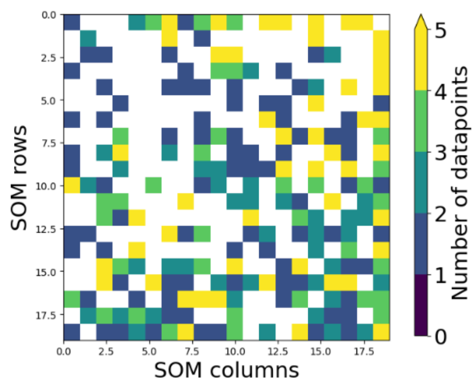

***Figure 25**: SOM After Unsupervised Training*

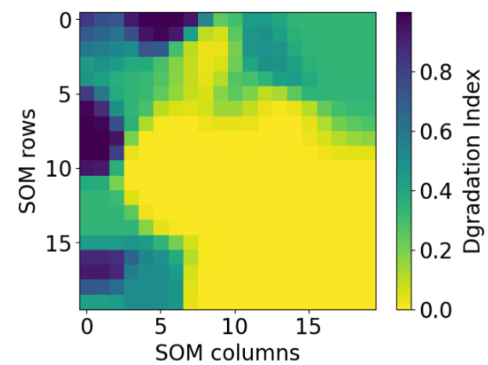

***Figure 26**: SOM After Supervised Training*

Compared to the results obtained in fully supervised learning, the map in figure 25 (to the left), is more populated explaining the more accurate predictions compared to the fully supervised run.

Up to five data points can share the same BMU. This can be beneficial for clustering purposes, but may also lead to a loss of model accuracy as they will be assigned the same predicted value. Fortunately, figure 26 (to the right) reveals that the BMUs with a higher number of data points are those with small degradation output, which is below the threshold of degradation. This suggests that the model is not assigning the same predicted value to data points with significantly different or high degradation indexes.

While nodes with a lower outcome are grouped together, there are three distinct clusters for nodes with higher degradation index exactly as in the fully supervised run. The following section 5.4.3.3. will give more insights on each of these clusters.

5.4.3.3. Cluster Analysis

Not to be confused with cluster, A, B and C of the previous run, the clusters of this run will be called D, E, and F respectively as shown in figure 27 below:

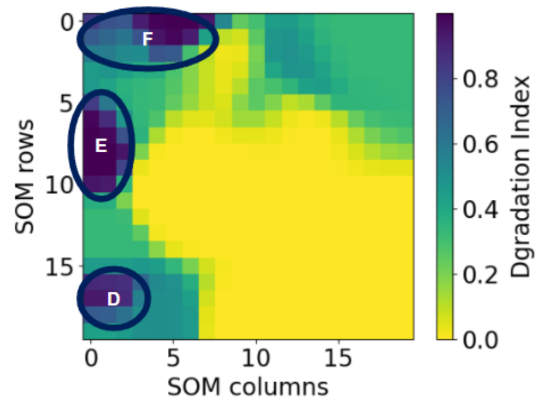

***Figure 27**: SOM Clusters in Semi-Supervised Run*

The feature importances plot of cluster D is in figure 28 below.

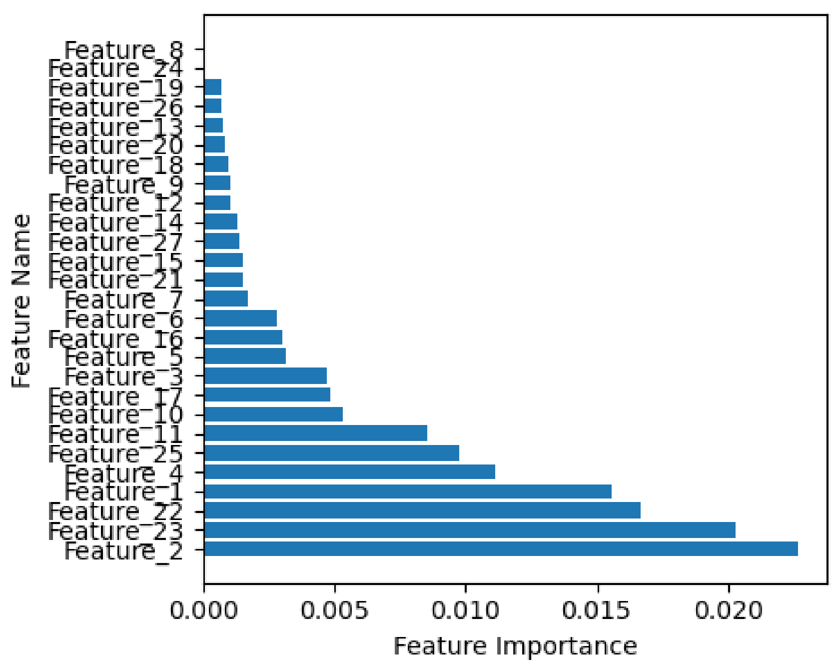

***Figure 28**: Feature Importances for Cluster D*

Features 2, 23, 22 and 1 are the main drivers for the prediction followed by features 4, 25 and 11. Feature 2 along with feature 1 are related to the fabrication of the component and are very high for this cluster confirming the hypothesis of the engineers stated beforehand.

Features 23 and 22 are measures of the same physical unit but at different altitudes. Referring to the analysis conducted in the fully supervised run, it is known that a combination of features 11 and 23 is likely to favour degradation. The same conclusion can be deduced here as the values of these features are over the mean of the dataset for aircrafts in this cluster. Feature 4 is the flight hours of the aircraft and is relatively high for this cluster explaining the high exposure to environmental conditions at feature 22 and 23 favouring the degradation.

Figure 29 below shows the feature effects for cluster E.

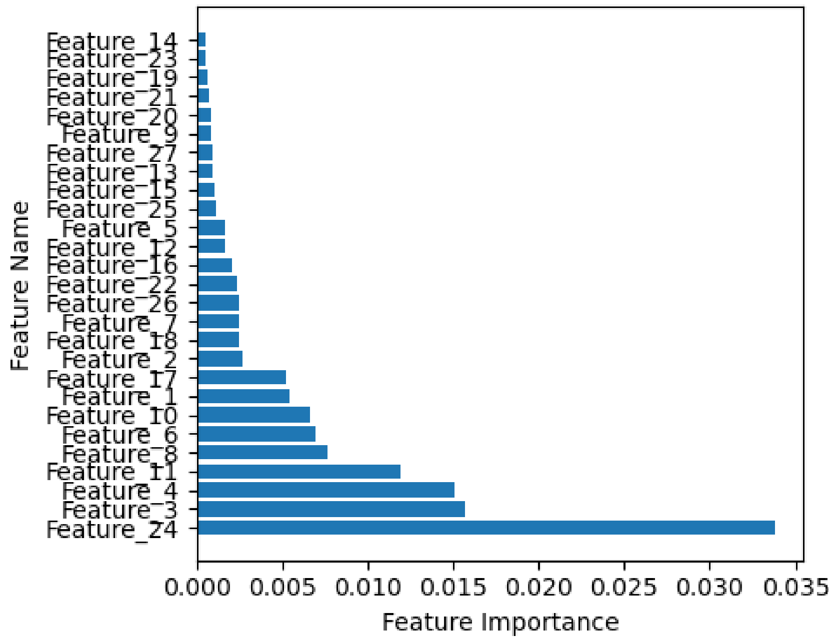

Figure 29: *Feature Importances for Cluster E*

For this cluster, feature 24 is the main contributor to the model's output followed by features 3 and 4, and 11. This is interesting as features 3 and 4 are respectively the flight cycles and flight hours of the aircraft and feature 24 measures an environmental condition cumulatively over the days when it is typically at its maximum.

This cluster contains aircrafts with the highest flight cycles and flight hours in the dataset; both maximum values are in this cluster (not the same aircraft though). These aircrafts are also among the oldest in the dataset even though the fleet remains rather young. As these aircrafts have flown the most, they have been exposed to the conduction measured in feature 24 the most which makes them prone to degradation.

Feature 11 is a cumulative measure of another environmental condition since the first flight of the aircraft. The cluster also contains the datapoint with the maximum values over the dataset and the overall values are all higher than the mean over the whole dataset. This could indicate that an excessive exposure to this condition is likely to cause degradation but not as likely as the one measured by feature 24.

Figure 30 below shows the feature effects for cluster F.

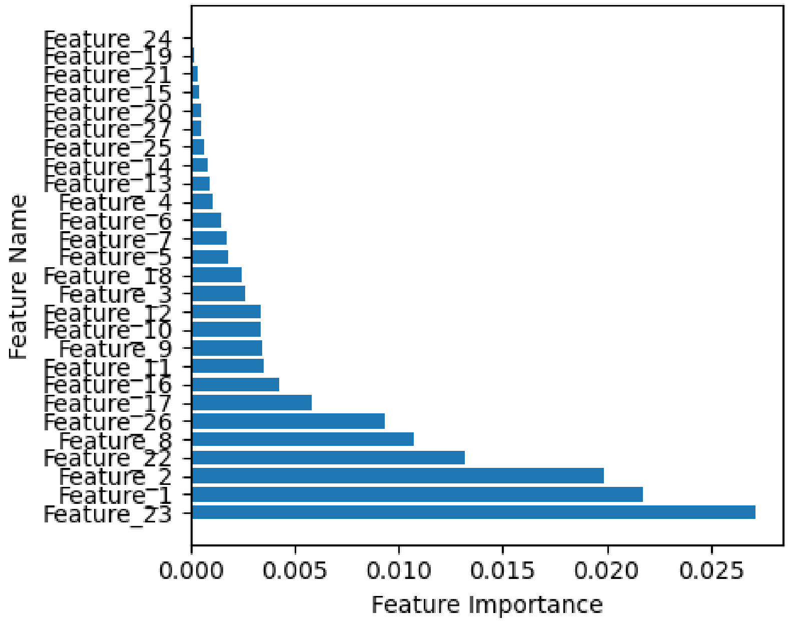

Figure 30*: Feature Importances for Cluster F*

Features 23, 1, and 2 are the main drivers for the degradation followed by features 22, 8, and 26. This cluster contains aircrafts that were parts of clusters B and C in the fully supervised run for which feature 23 was also the main contributor. Features 1 and 2 are related to the manufacturing of the component and high values, which is the case here, are predicted to cause sensitivity of the component to certain environmental conditions.

This cluster contains aircrafts that are similar to those in cluster D, with the only difference being that the values of feature 23 are higher for this cluster which explains the order of feature importances for this cluster.

In contrast to the fully supervised run, where almost the same drivers were present across all clusters with varying amplitudes, the semi-supervised run reveals distinct drivers for each cluster. This suggests that the semi-supervised approach is better at capturing the patterns and relationships within each cluster, which was the primary objective of this analysis. This explains why this SOM has a better performance.

5.4.4. Reference Models

To provide a broader context for evaluating the performance of the self-organizing map, this section presents a comparative analysis with several existing models in the field, as a benchmark for SOM's performance.

In this comparison, four models will be examined: XGBoost, which laid the groundwork for this project; Random Forest, serving as the chosen baseline; the simple yet informative Dummy

Regressor from scikit-learn; and a Symbolic Regressor, known for its effectiveness on small datasets [39].

The symbolic regression used is the one introduced by Jin, Y et al. [40]. It employs Bayesian approach to symbolic regression, formulating the problem within a Bayesian framework, using a Markov chain Monte Carlo sampling to explore the space of possible expressions, and selecting the most likely expression from the posterior distribution, allowing for principled uncertainty quantification and incorporation of prior knowledge [40].

Table 10 below details the performance of each of these models in train and test :

Model	Train MAE	Train RMSE	Test MAE	Test RMSE
Dummy Regression	0.282	0.364	0.311	0.378
XGBoost	0.147	0.184	0.175	0.212
Random Forest	0.094	0.158	0.167	0.187
Bayesian Symbolic Regression	0.118	0.141	0.127	0.142
Supervised SOM	0.058	0.100	0.102	0.111
Semi-Supervised SOM	0.050	0.097	0.061	0.084

***Table 10**: Performance Metrics of All Models*

As expected, the Dummy Regression model performs poorly, with high MAE and RMSE values. This is because it's a simple baseline model that doesn't take into account any underlying relationships in the data. This could be understood as the minimal performance to be expected of a model. All models outperform the Dummy Regressor which is a good sign.

Surprisingly, Random Forest outperforms XGBoost here. This is probably because Random Forest can be more robust to overfitting, when dealing with small or high-dimensional data. This is because Random Forest combines multiple decision trees , which helps to reduce overfitting by averaging out the errors, whereas XGBoost might require larger datasets to achieve optimal performance.

The performance of the Bayesian Symbolic Regression model is interesting, as it falls somewhere in the middle of the pack. This suggests that it may be a useful model in certain situations, but it's not necessarily the best choice for this particular problem.

The Supervised SOM and Semi-Supervised SOM models clearly outperform the other models, with significantly lower MAE and RMSE values. This means that the Self-Organizing Map approach is well-suited to this problem, and that the semi-supervised variant may be particularly effective.

5.4. Conclusion & Future Work

This second study addresses the estimation of the degradation of an aircraft component through Self-Organizing Map (SOM), a typically unsupervised learning approach. The methodology to building training, and interpreting SOM has been detailed. Two runs of the SOM were performed, with the semi-supervised variant outperforming all other models. This is likely due to the fact that the semi-supervised SOM employs all data points, allowing it to better capture correlations between features.

The result of the modelling indicates that there are three main features that influence the degradation of the studied component. At a first glance, the Airbus Zero AOG teams were surprised by one of these features, as they did not expect it to play a determining role. Their next step will be to liaise with specialised experts to further investigate how exactly the feature affects the physical properties of the component.

In the meantime however, the model does identify the correct aircraft that will have a degraded component. Thus, it is useful and actionable already and it will be used to predict the degree of component degradation. From Airbus' perspective, this will provide proactive maintenance planning and the associated amount of required supplies/parts to be purchased. From customers' perspective, it will reduce the likelihood of unexpected failures and minimise downtime, thus resulting in an improved customer experience.

Finally, the ultimate desired outcome of this study is to understand the root cause of the component degradation and potentially eliminate it completely by improving the design of the component.

6. Conclusion

This report presents two examples of using unsupervised learning approaches for regression-like modelling; for market segmentation and predictive maintenance.

In the first study, Fuzzy C-Means was used to estimate the digital maturity of airlines. Although the initial goal was to estimate the digital maturity of all airlines around the world, the scope was reduced to airlines using the Skywise Platform due to challenges in retrieving necessary data. The proposed method begins by a dimensionality reduction followed by a classical c-means clustering. Following that the clusters will be ordered using expert-based rules as per their digital status and the membership matrix of the c-means will be used to estimate digital maturity. A dashboard demonstrating the key insights retrieved from this method was developed and deployed on Skywise.

This study was completed by a second study employing Self-Organizing Map (SOM) for predictive maintenance. A methodology on how to build and configure the SOM was detailed. This methodology trains the SOM in two steps, an unsupervised learning phase where a Best Matching Unit (BMU) is assigned to each datapoint and a supervised learning phase where an output is predicted based on the calculated BMU. Two runs of the SOM were performed, with the semi-supervised variant outperforming the fully supervised one.

In conclusion, this report demonstrates the potential of unsupervised learning approaches in addressing complex problems in different domains. The next steps will involve collecting more target values to test the generalisation of the models on bigger datasets.

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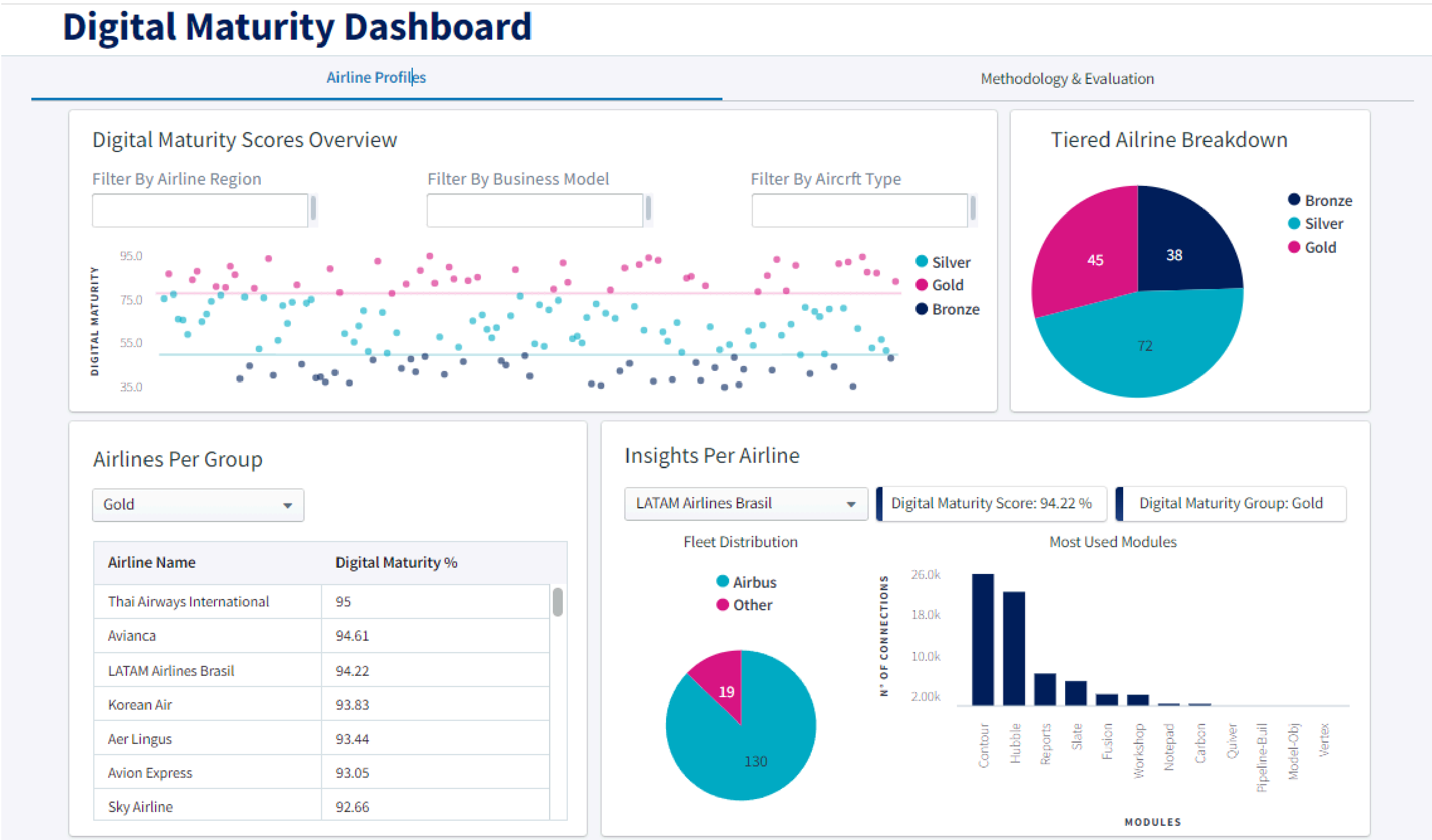
Appendix A - Digital Maturity Dashboard | Airline Profiles



Appendix B - Digital Maturity Dashboard | Methodology & Evaluation

Digital Maturity Dashboard

Airline Profiles | Methodology & Evaluation

Methodology

Objective

This dashboard aims to provide a data-driven assessment of airline digital maturity, analysing engagement patterns on Skywise.

Data Sources

The data used in this dashboard is sourced from skywise usage metrics for each airline over the past year, i.e. the number of unique accounts, the average number of connections per account and per day, and the average session length for each module available on skywise, alongside fleet information (age and distribution).

Step-by-Step Breakdown

To synthesize this data into a unified digital maturity index, we constructed a multi-layered framework incorporating:

- **Feature Prioritisation:** Following data scaling, a weighted importance score is assigned to each feature within the dataset.
- **Dimensionality Reduction:** The number of features is reduced, making it easier to analyse and visualise while retaining the most important information.
- **Clustering:** A machine learning algorithm is employed, capturing the degree of association between each airline and its corresponding digital maturity groups.
- **Cluster Ordering:** Through the application of expert-based rules, the identified clusters are subsequently ordered from most to least digitally mature.
- **Digital Maturity Score Calculation:** By leveraging the memberships to digital groups, a score is calculated for each airline, providing a quantifiable measure of their digital maturity.

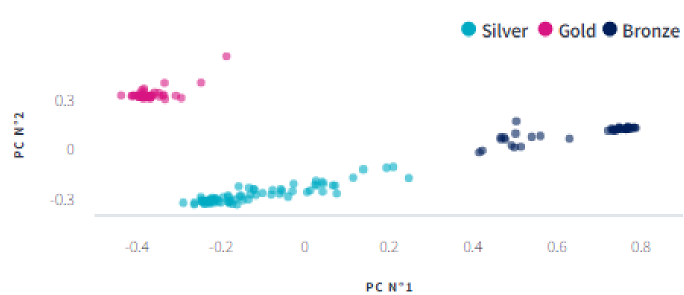
Model Monitoring

Silhouette Score
0.811

Conserved Variance
84.96%

Date of Last Update
2024-09-09

Clusters Visualization



Feature Importances